Document 09

Ambiguity Register

1. Purpose of This Document

This document records canon relevant ambiguities, conflicts, rejected assumptions, and formal resolutions discovered during the preparation of the Trinity Engine v1 canon package.

Its purpose is to:

- expose ambiguity instead of hiding it
- prevent informal prior assumptions from re entering implementation
- record why certain canon choices were made
- track conflicts between older materials and the final canon
- support disciplined canon change control

This document is a control instrument.

It does **not** replace the governing canon documents.
Any resolved ambiguity recorded here must be folded back into the authoritative canon document or documents that govern the affected behavior.

---

2. Register Use Rules

2.1 What Must Be Recorded

An item must be recorded in this register if it involves any of the following:

- conflicting source material
- ambiguous implementation behavior
- prior assumptions that were rejected
- prototype logic that differs from final canon
- unclear scope boundaries
- unclear source authority
- unresolved determinism risks
- temporarily accepted working assumptions pending later confirmation

2.2 What This Register Does Not Do

This register must not be used as a substitute for:

- updating Document 04 for input behavior

- updating Document 05 for calculation behavior
- updating Document 06 for mapping assets
- updating Document 07 for output behavior
- updating Document 08 for versioning and governance behavior

If an ambiguity is resolved, the authoritative canon documents must still be updated.

2.3 Status Values

Each register item must use one of the following statuses:

- `open`
- `resolved`
- `rejected`
- `folded_back`

Meaning of statuses

- `open` = ambiguity exists and is not yet resolved
- `resolved` = decision has been made, but not yet fully folded back everywhere required
- `rejected` = prior assumption or prior candidate has been explicitly rejected
- `folded_back` = the resolution has been fully reflected into the governing canon documents

---

3. Register Structure

Each ambiguity item must contain:

- `item_id`
- `title`
- `status`
- `affected_documents`
- `ambiguity_description`
- `resolution`
- `rejected_alternatives`
- `notes`

---

4. Ambiguity Items

Item A01

item_id: `A01`
title: Engine input boundary versus broader system intake pipeline
status: `folded_back`
affected_documents:

- Document 01
- Document 04
- Document 09

ambiguity_description:
Older materials described a broader system flow in which birth input handling included birthplace parsing, coordinate assignment, and timezone determination. This created ambiguity about whether Trinity Engine itself should perform those intake normalization steps.

resolution:
Trinity Engine v1 receives normalized, validated input only.
Raw birthplace search, geocoding, place disambiguation, and timezone determination are outside engine scope.

rejected_alternatives:

- engine accepts raw birthplace strings
- engine performs place resolution internally
- engine accepts ambiguous timezone inputs

notes:
This resolution has been folded back into the governing canon documents.

Item A02

item_id: `A02`
title: Use of PoC canon versus master calculation specification as source of truth
status: `folded_back`
affected_documents:

- Document 02
- Document 03
- Document 05
- Document 06
- Document 09

ambiguity_description:
Earlier discussion referred informally to “the same objects from the PoC canon,” which created a risk that prior shared memory would substitute for explicit canon documentation.

resolution:
The master calculation specification is treated as the source of truth for supported objects and calculation foundations, unless the final canon package explicitly revises or narrows that material.

rejected_alternatives:

- relying on prior chat memory
- relying on PoC assumptions without explicit restatement
- mixing prototype and final canon sources informally

notes:
This resolution has been folded back through the explicit supported objects, source stack, calculation, and mapping documents.

Item A03

item_id: `A03`
title: Separate Trinity derived calculations category
status: `folded_back`
affected_documents:

- Document 01
- Document 02
- Document 09

ambiguity_description:
Earlier draft language referred to “Trinity derived calculations” as if it were its own supported category, but that category was not concretely defined.

resolution:
There is no separate supported category called Trinity derived calculations in v1.
All supported outputs must belong explicitly to Astrology, Human Design, or Gene Keys.

rejected_alternatives:

- keeping a vague fourth category
- allowing derived outputs with no domain assignment

notes:
This ambiguity has been fully folded back.

Item A04

item_id: `A04`
title: Human Design scope too thin versus credible v1 scope
status: `folded_back`
affected_documents:

- Document 02
- Document 05
- Document 06
- Document 07
- Document 09

ambiguity_description:
An earlier conservative scope limited Human Design mainly to activations, Type, Authority, Profile, and Incarnation Cross. This raised the question of whether v1 would be structurally incomplete relative to standard Human Design expectations.

resolution:
Human Design v1 includes:

- Personality activations
- Design activations
- Channels
- Centers
- Definition
- Type
- Authority
- Profile
- Incarnation Cross

rejected_alternatives:

- excluding Channels
- excluding Centers
- excluding Definition while still claiming a structurally complete Human Design layer

notes:
This resolution has now been folded back into scope, calculation, mapping, and output documents.

Item A05

item_id: `A05`
title: Gene Keys identifiers versus Gene Keys content layer

status: `folded_back`
affected_documents:

- Document 02
- Document 05
- Document 06
- Document 07
- Document 09

ambiguity_description:
It was unclear whether Gene Keys v1 should include only identifiers and Activation Sequence positions, or also include shadow, gift, and essence text.

resolution:
Gene Keys v1 includes identifiers only:

- key
- line
- Activation Sequence positions

Shadow, gift, and essence text remain outside engine scope and belong to the content layer.

rejected_alternatives:

- including Gene Keys prose in engine output
- embedding semantic state meaning inside the calculation layer

notes:
This ambiguity has been fully folded back.

Item A06

item_id: `A06`
title: Strict numeric typing at engine boundary
status: `folded_back`
affected_documents:

- Document 04
- Document 09

ambiguity_description:
It was unclear whether numeric fields such as latitude and longitude should accept numeric looking strings and be coerced internally.

resolution:
Numeric fields must be submitted as true numeric values.
Numeric looking strings are invalid and must be rejected as `InvalidInput`.

rejected_alternatives:

- silent coercion of `"52.0116"` into `52.0116`
- mixed type acceptance at engine boundary

notes:
This ambiguity has been fully folded back into the input contract.

Item A07

item_id: `A07`
title: Old Gate 1 anchor assumption
status: `folded_back`
affected_documents:

- Document 05
- Document 06
- Document 09

ambiguity_description:
An earlier assumption stated that Gate 1 Line 1 begins at `313.25°`. This conflicted with Human Design transit positioning logic and the annual cycle around the Rave New Year.

resolution:
The `313.25°` Gate 1 anchor assumption is rejected.

rejected_alternatives:

- continuing to use `313.25°` as the authoritative Gate 1 Line 1 anchor
- building the final gate sequence on top of the rejected anchor

notes:
This rejection has been folded back into the active canon by removing superseded anchor logic and replacing it with the adopted canonical anchor basis.

Item A08

item_id: `A08`
title: Canonical gate sequence anchor
status: `folded_back`
affected_documents:

- Document 05
- Document 06

- Document 09

ambiguity_description:
After rejecting the earlier Gate 1 anchor assumption, a correct angular anchor was required for the pinned 64 gate sequence.

resolution:
The canonical Human Design gate sequence is anchored as follows:

- Gate 38 begins at `277.5°`
- Gate 41 begins at `300.0°`

This aligns the Rave New Year with Gate 41 in late January and makes the pinned 64 gate sequence internally coherent.

rejected_alternatives:

- retaining the old Gate 1 based anchor
- leaving the gate sequence unanchored
- using an inferred anchor without consistency to the annual transit cycle

notes:
This resolution has been folded back into the active calculation and mapping canon.

Item A09

item_id: `A09`
title: Human Design structural rules not explicitly pinned in master calculation specification
status: `folded_back`
affected_documents:

- Document 05
- Document 06
- Document 09

ambiguity_description:
The master calculation specification names Type, Authority, Profile, and Incarnation Cross, but does not fully formalize centers, channels, definition, Type derivation, or Authority derivation rules in explicit canon language.

resolution:
The final canon package explicitly adopts deterministic structural rules for:

- channels
- centers
- definition
- Type

- Authority
- Incarnation Cross structural encoding

rejected_alternatives:

- relying on “standard Human Design rules” without explicit canon wording
- allowing the engineer to infer BodyGraph mechanics from general domain knowledge

notes:
This ambiguity has now been folded back into the governing calculation and mapping documents.

Item A10

item_id: `A10`
title: Incarnation Cross naming versus structural encoding
status: `folded_back`
affected_documents:

- Document 05
- Document 06
- Document 07
- Document 09

ambiguity_description:
It was unclear whether Incarnation Cross output required a human readable cross name or whether structural encoding alone was sufficient.

resolution:
For v1, Incarnation Cross is output as structural encoding only:

- Personality Sun gate and line
- Personality Earth gate and line
- Design Sun gate and line
- Design Earth gate and line

A human readable cross name is not required unless a naming table is explicitly pinned later.

rejected_alternatives:

- inventing or assuming a naming table without pinned authority
- outputting partial cross information only

notes:
This ambiguity has been fully folded back.

Item A11

item_id: `A11`
title: Hashes in canonical output metadata
status: `folded_back`
affected_documents:

- Document 01
- Document 07
- Document 09

ambiguity_description:
It was unclear whether normalized input hashes and output hashes should be included in canonical output metadata for v1.

resolution:
Hashes are postponed from canonical output in v1.

rejected_alternatives:

- including hashes immediately in canonical output
- treating operational hash logging as part of the engine contract

notes:
This ambiguity has been folded back into boundary and output rules.

Item A12

item_id: `A12`
title: Multi canon version support versus single pinned canon for v1
status: `folded_back`
affected_documents:

- Document 01
- Document 08
- Document 09

ambiguity_description:
It was unclear whether Trinity Engine v1 should support multiple canon versions from the start.

resolution:
Trinity Engine v1 supports one pinned canon version only.

rejected_alternatives:

- multi canon support in v1

- runtime selectable canon versions
- backward compatibility shims in v1

notes:
This ambiguity has been fully folded back.

Item A13

item_id: `A13`
title: Longitude precision and design time output format
status: `folded_back`
affected_documents:

- Document 07
- Document 09

ambiguity_description:
The output contract originally left open the numeric precision for longitude outputs and the exact datetime format for `design_time_utc`.

resolution:
Pinned output formatting:

- longitude values are numeric and rounded to 6 decimal places
- `design_time_utc` uses ISO 8601 UTC format with trailing `Z`
- `design_time_utc` uses whole second precision

rejected_alternatives:

- leaving precision unspecified
- using variable datetime formatting
- using millisecond precision by default

notes:
This ambiguity has been fully folded back into the output specification.

Item A14

`item_id:` A14
`title:` Timezone database family pinned but exact release not yet canonized
`status:` folded_back
`affected_documents:`

- Document 03
- Document 04

- Document 08
- Document 09

`ambiguity_description:`
The canon requires timezone aware UTC conversion, but the exact timezone database release was not explicitly pinned. This left open the possibility of cross environment differences in historical timezone and daylight saving behavior.

`resolution:`
Trinity Engine v1 must use one explicitly pinned timezone database release as the authoritative timezone reference for canonical UTC conversion. The governing source stack and versioning documents must record that pinned release explicitly.

`rejected_alternatives:`

- treating the IANA timezone database family as sufficiently specific without a pinned release
- allowing environment default timezone data to govern canonical conversion
- accepting runtime timezone data differences as harmless

`notes:`
This item should remain `resolved` until the exact timezone database release is visibly inserted into the governing documents.

Item A15

`item_id:` A15
`title:` House interval ownership rule missing from governing calculation canon
`status:` folded_back
`affected_documents:`

- Document 05
- Document 06
- Document 09

`ambiguity_description:`
Astrology house placement was defined as requiring canonical house cusp structure, but the canon did not explicitly state how cusp boundary ownership must be resolved for objects exactly on a house boundary.

`resolution:`
For Trinity Engine v1, each house interval begins at its own cusp inclusive and ends at the next cusp exclusive. A longitude exactly equal to a house cusp belongs to the house beginning at that cusp. Wraparound between House 12 and House 1 follows the same rule.

`rejected_alternatives:`

- leaving house boundary ownership to external library defaults
- treating house cusp ownership as implicitly understood
- allowing dual ownership or ambiguous boundary behavior

`notes:`
This item should move to `folded_back` only after the rule is inserted into Document 05 and reflected by reference in Document 06.

Item A16

`item_id:` A16
`title:` Design time final timestamp selection under sub second stop condition
`status:` folded_back
`affected_documents:`

- Document 05
- Document 07
- Document 09
- Document 10

`ambiguity_description:`
The canon defined Design time search by bisection and specified the stop conditions, but did not explicitly state which timestamp is canonical when the final search interval becomes narrower than one second.

`resolution:`
If the Design time search stops because the remaining interval width is less than 1 second, the canonical Design time is the lower bound of the final search interval. The emitted `design_time_utc` value uses whole second precision by truncation to that lower bound.

`rejected_alternatives:`

- leaving final timestamp selection implied by the search process
- using nearest second rounding without explicit canon adoption
- using midpoint or upper bound selection without formal canon wording

`notes:`
This item should move to `folded_back` only after the governing calculation, output, and golden test documents visibly reflect this rule.

Item A17

`item_id:` A17
`title:` Error fixture exactness depended on free text message wording without a pinned message catalog

`status:` folded_back
`affected_documents:`

- Document 07
- Document 09
- Document 10

`ambiguity_description:`
The output and golden fixture documents encouraged exact error matching, but no authoritative catalog of canonical message strings was pinned. This made free text developer messages appear canon critical without a real canon basis.

`resolution:`
Canonical error comparison for Trinity Engine v1 uses `error_type` and `error_code` as authoritative fields. The `message` field remains required developer facing text, but its wording is not authoritative for fixture exactness unless a later canon revision pins a formal message catalog.

`rejected_alternatives:`

- treating free text message wording as exact canonical truth without a catalog
- relying only on coarse error types without specific error codes
- leaving error comparison semantics implicit

`notes:`
This item should move to `folded_back` only after Document 07 and Document 10 visibly align on this rule.

Item A18

`item_id:` A18
`title:` Error classification boundaries between invalid, incomplete, and unsupported input were under specified
`status:` folded_back
`affected_documents:`

- Document 07
- Document 09

`ambiguity_description:`
The canon named allowed error types, but did not sufficiently operationalize how to classify edge cases such as incomplete time fields, timezone abbreviations, or structurally valid but unsupported identifiers.

`resolution:`
Document 07 must include explicit classification rules and examples distinguishing:

- `invalid_input`
- `incomplete_input`
- `unsupported_input`

`rejected_alternatives:`

- relying on general intuition about neighboring error classes
- leaving classification behavior to implementation judgment
- treating all boundary failures as invalid input

`notes:`
This item should move to `folded_back` only after the classification matrix is inserted into Document 07.

Item A19

`item_id:` A19
`title:` Accepted timezone identifier forms were not explicitly narrowed
`status:` folded_back
`affected_documents:`

- Document 04
- Document 07
- Document 09

`ambiguity_description:`
The canon required timezone based input and timezone aware conversion, but it did not explicitly define whether abbreviations, aliases, and non canonical identifier forms were accepted at the engine boundary.

`resolution:`
For Trinity Engine v1, accepted timezone identifiers are restricted to canonical IANA `Area/Location` names. Abbreviations, aliases, and link names are not accepted unless explicitly introduced by later canon revision.

`rejected_alternatives:`

- accepting timezone abbreviations such as CET or EST
- accepting alias forms such as US/Eastern by default
- silently normalizing non canonical timezone identifiers inside the engine

`notes:`
This item should remain `resolved` until the input and output documents visibly reflect the accepted identifier rule.

Item A20

`item_id:` A20

`title:` Success response exactness language overstated serializer level precision

`status:` folded_back

`affected_documents:`

- Document 07
- Document 09
- Document 10

`ambiguity_description:`

The canon used strict exactness language for success outputs, but did not explicitly distinguish semantic JSON equality from literal byte for byte JSON serialization. This created false precision where serializer behavior and object member ordering were not themselves canonized.

`resolution:`

Canonical success response comparison for Trinity Engine v1 uses semantic JSON equality against the output contract. Exact scalar values, canonical field names, and canon pinned ordered arrays must match, but literal byte for byte object serialization is not required unless later canon explicitly pins serializer behavior and object member ordering.

`rejected_alternatives:`

- implying byte for byte equality without pinning serializer behavior
- weakening comparison to loose logical similarity
- allowing unordered collections where canonical order is explicitly required

`notes:`

This item should move to `folded_back` only after Document 07 and Document 10 visibly align on semantic success comparison.

Item A21

`item_id:` A21

`title:` First golden fixture freeze risked circular self validation

`status:` folded_back

`affected_documents:`

- Document 09
- Document 10
- Document 12

`ambiguity_description:`

The golden test specification required real fixture population from a canon compliant engine, but did not yet make explicit that the first approved fixture set requires review beyond initial

engine generation. This left open the risk that the first fixture oracle could become self validating.

`resolution:`
The first approved fixture set for Trinity Engine v1 must not be frozen solely because it was generated by a canon compliant engine. Candidate fixtures must also undergo review against the governing canon, including independent review of ambiguity prone boundary areas, before being marked approved.

`rejected_alternatives:`

- freezing the first fixture set immediately after initial engine generation
- relying on implementation output alone as sufficient oracle creation
- treating fixture approval as implicit rather than explicit

`notes:`
This item should move to `folded_back` only after Document 10 visibly includes fixture approval and freeze requirements.

---

5. Open Items

Item O01

item_id: `O01`
title: Exact Swiss Ephemeris version pin
status: `folded_back`
affected_documents:

- Document 03
- Document 08
- Document 09

ambiguity_description:
Swiss Ephemeris is pinned as the authoritative astronomical source for Trinity Engine v1, but the exact version number was not initially inserted into the canon package.

resolution:
The authoritative Swiss Ephemeris version for Trinity Engine v1 is:

- `2.10.03`

This version is canon pinned and must be used consistently across implementation, testing, and review for v1.

rejected_alternatives:

- leaving the Swiss Ephemeris version unspecified
- allowing implementer choice of Swiss Ephemeris version
- treating “latest available version” as acceptable for v1

notes:
This ambiguity has been folded back into:

- Document 03. Canonical Source Stack
- Document 08. Determinism and Versioning Rules

Item O02

`item_id:` O02
`title:` Final closure pass across governing, output, and golden test documents
`status:` folded_back
`affected_documents:`

- Document 05
- Document 06
- Document 07
- Document 09
- Document 10
- Document 12
- Document 00

`ambiguity_description:`
The canon package is substantially formalized, but a final closure pass is still required to ensure that newly resolved ambiguity items from the Pascal review are fully folded back into the governing canon, reflected in acceptance rules, recorded in governance documents, and marked consistently in package status tracking.

This closure pass must ensure that:

- governing rule homes contain the operative text
- secondary documents reference governing rule homes without duplicating or drifting
- output rules and fixture rules align on comparison semantics
- error handling semantics align across output and test documents
- package status language does not overstate finality while known closure items remain unresolved

`resolution:`
The package will complete a targeted fold back pass covering:

- house interval ownership
- Design time final timestamp emission

- error comparison semantics
- error classification matrix
- timezone identifier acceptance policy
- success comparison semantics
- first fixture freeze governance
- timezone database release pin
- final status alignment in Document 00

`rejected_alternatives:`

- treating near final documents as final before the new ambiguity items are folded back
- correcting only rationale or control documents without updating governing rule homes
- declaring package closure before acceptance semantics and tracker status align

`notes:`

This item should remain `open` until the full Pascal closure set is visibly inserted into the relevant documents and the final package tracker is updated. It should move first to `resolved` when the fold back decisions are complete in the working drafts, and only then to `folded_back` when the final package reflects those decisions consistently.

---

6. Fold Back Rule

A resolved ambiguity item may be marked `folded_back` only when:

- the resolution has been inserted into the governing canon documents
- any earlier conflicting wording has been removed or superseded
- no canon critical dependency remains open for that item

A resolution discussed in chat but not reflected in the governing documents is not considered folded back.

A register item identified during external implementation review must not be marked `folded_back` merely because a resolution was agreed in discussion. Visible insertion into the governing package remains required.